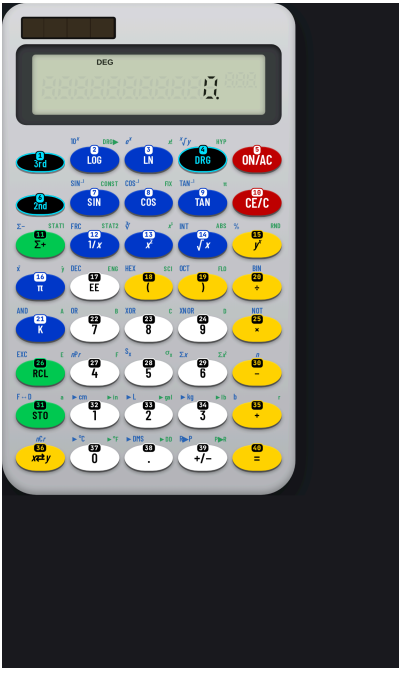
U-45 CALCULATORS

User Manual

Calculator #12 One-Line Scientific with 3rd/2nd (solar, silver)

Scientific – classic one-line



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#12 — One-Line Scientific with 3rd/2nd (solar, silver)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example 

. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

AOS algebraic (operator precedence, = completes).

Chapters

1. Getting started
2. Features
3. Numbered key map
4. Key-by-key reference
5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
9. Practice drills
10. Appendix A — Drill answers
11. Appendix B — Function finder (A–Z)
12. Appendix C — Glossary

13. Appendix D — Model notes & limits

14. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#12** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#17, #22, #23, #24, #27, #28, #29, #32, #33, #34, #37, #38, #39
op	Operators & execute	#15, #18, #19, #20, #25, #30, #35, #36, #40
fn	Functions	#2, #3, #7, #8, #9, #12, #13, #14, #16, #21
mem	Memory, variables & registers	#11, #26, #31
clr	Clear, delete & power	#5, #10
mod	Shift, modes, menus & navigation	#1, #4, #6

2 · Features

67 features: 48 live on the keys, 9 app/UI features, 10 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE One-line AOS with 2nd and 3rd shifts
2. LIVE STAT1 / STAT2 with Σ+ / Σ− data entry (x⇄y separates pairs)
3. LIVE DEC / HEX / OCT / BIN with hex digits A–F on 3rd
4. LIVE Unit conversions ►cm/►in, ►L/►gal, ►kg/►lb, °F⇄°C
5. LIVE FIX, SCI, ENG, FLO display modes
6. LIVE Constant K operation
7. LIVE EXC memory exchange
8. LIVE Fraction template ▭/▭ with exact fraction results
9. LIVE S⇔D / F⇔D toggle between exact (fraction, √, π) and decimal answers
10. LIVE Square root $\sqrt{}$
11. LIVE Cube root $\sqrt[3]{}$
12. LIVE Any-index root $\sqrt[x]{y}$
13. LIVE Square x^2
14. LIVE Cube x^3
15. LIVE Reciprocal x^{-1} (1/x)
16. LIVE Power x^y / ^ with right-to-left chaining (2^3^2 = 512)
17. LIVE Common logarithm $\log_{10}$
18. LIVE Natural logarithm ln
19. LIVE Antilog 10^x
20. LIVE Exponential e^x

21. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
22. LIVE **Inverse trigonometry $\sin^{-1}$ / $\cos^{-1}$ / $\tan^{-1}$ with domain checking (Math ERROR outside −1…1)**
23. LIVE **Hyperbolic sinh / cosh / tanh and their inverses via hyp**
24. LIVE **Angle units DEG / RAD / GRA with unit-suffix conversion (°, r, g)**
25. LIVE **Sexagesimal ° ′ ″ entry and decimal ↔ DMS conversion**
26. LIVE **Rectangular → polar conversion Pol(x,y) → r, θ**
27. LIVE **Polar → rectangular conversion Rec(r,θ) → x, y**
28. LIVE **Factorial x! (exact up to 69!)**
29. LIVE **Permutations nPr**
30. LIVE **Combinations nCr**
31. LIVE **Rnd — round the value to the displayed precision**
32. LIVE **Absolute value |x|**
33. LIVE **Percent key (x %, add-on and discount chains)**
34. LIVE **π constant to full internal precision**
35. LIVE **Scientific-notation entry ×10^x / EXP / EEX**
36. LIVE **Sign change (−) / +/− / CHS**
37. LIVE **Engineering notation ENG (exponent multiple of 3)**
38. LIVE **Store a value into memory / a variable (STO, sto►, →, Min)**
39. LIVE **Recall memory / variables (RCL, MR, recall)**
40. LIVE **Statistics: data entry, n, $\bar{x}$, σ, s, Σx, Σx^2**
41. LIVE **Two-variable statistics and linear regression (a, b, r)**
42. LIVE **BASE-N: DEC / HEX / BIN / OCT conversions**
43. LIVE **Bitwise logic AND / OR / XOR / XNOR / NOT**

44. LIVE Built-in scientific constants
45. LIVE Unit conversions (length, mass, volume, temperature)
46. LIVE Exchange x ⇄ y / x ⇄ M
47. LIVE Constant calculations (K) on repeated operators
48. LIVE FIX — fixed number of decimal places
49. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
50. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
51. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
52. APP/UI Show / hide key numbers with one button (remembered between sessions)
53. APP/UI Copy the current display text to the clipboard
54. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
55. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
56. APP/UI Reference photo viewer and tool-tip name on every key
57. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
58. PLANNED Two-line display preview
59. PLANNED Fraction simplify step
60. PLANNED Complex mode
61. PLANNED Regression ŷ / x̂ prediction
62. PLANNED Physical constants list

63. PLANNED **Answer history export**

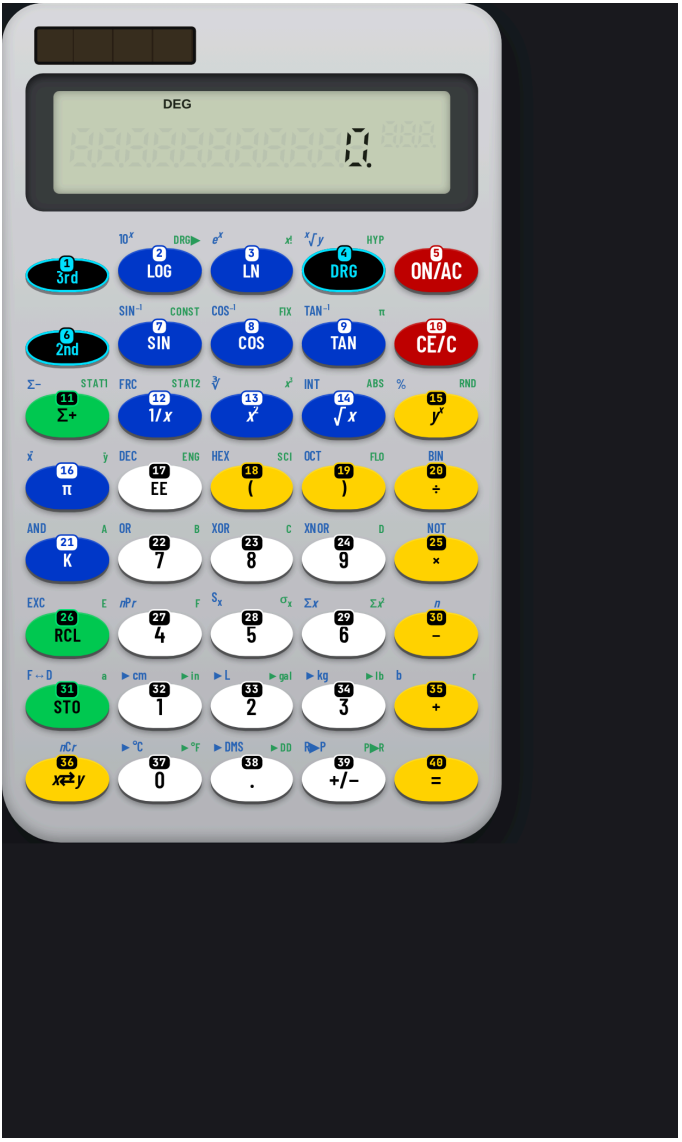
64. PLANNED **Speech read-out of results**

65. PLANNED **Colour-blind key theme**

66. PLANNED **Exam lock**

67. PLANNED **Custom shortcut keys**

3 · Numbered key map



Calculator #12 with key numbers shown (Hi-Contrast colours).

Key table

"Shift" = 2nd / f / SHIFT layer; "Alpha" = ALPHA layer; "3rd" = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	3rd	·	·	·	mod
2	LOG	10x	·	DRG▶	fn
3	LN	ex	·	x!	fn
4	DRG	x√y	·	HYP	mod
5	ON/AC	·	·	·	clr
6	2nd	·	·	·	mod
7	SIN	SIN^{-1}	·	CONST	fn
8	COS	COS^{-1}	·	FIX	fn
9	TAN	TAN^{-1}	·	π	fn
10	CE/C	·	·	·	clr
11	Σ+	Σ−	·	STAT1	mem
12	1/x	FRC	·	STAT2	fn
13	x^2	$\sqrt[3]{}$	·	x^3	fn
14	√x	INT	·	ABS	fn
15	yx	%	·	RND	op
16	π	$\bar{x}$	·	$\bar{y}$	fn
17	EE	DEC	·	ENG	num
18	(	HEX	·	SCI	op
19	)	OCT	·	FLO	op
20	÷	BIN	·	·	op
21	K	AND	·	A	fn
22	7	OR	·	B	num
23	8	XOR	·	C	num

#	Face	Shift	Alpha	3rd	Group
24	9	XNOR	·	D	num
25	×	NOT	·	·	op
26	RCL	EXC	·	E	mem
27	4	nPr	·	F	num
28	5	Sx	·	σx	num
29	6	Σx	·	Σx^2	num
30	−	n	·	·	op
31	STO	F ↔ D	·	a	mem
32	1	▶ cm	·	▶ in	num
33	2	▶ L	·	▶ gal	num
34	3	▶ kg	·	▶ lb	num
35	+	b	·	r	op
36	x⇄y	nCr	·	·	op
37	0	▶ °C	·	▶ °F	num
38	.	▶ DMS	·	▶ DD	num
39	+/−	R▶P	·	P▶R	num
40	=	·	·	·	op

4 · Key-by-key reference

Every key on calculator #12, in number order, with every function it carries in this app.

#1 3rd `Shift, modes, menus & navigation`

Primary: 3rd — the next key uses its third function

#2 LOG `Functions`

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #6 first): 10x — 10^x — antilog

3rd / g layer (press #1 first): DRG▶ — DRG▶ — convert the displayed angle to the next unit

Keyboard: `l`

#3 LN `Functions`

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #6 first): ex — e^x — exponential

3rd / g layer (press #1 first): x! — Factorial n! (also on non-integers via the gamma function where supported)

Keyboard: `n`

#4 DRG `Shift, modes, menus & navigation`

Primary: DRG — choose/convert the angle unit (Deg / Rad / Grad)

shifted (2nd / f / SHIFT layer) (press #6 first): $x\sqrt{}y$ — x-th root of y

3rd / g layer (press #1 first): HYP — Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh

#5 ON/AC `Clear, delete & power`

Primary: ON / AC — power on, or clear all

#6 2nd `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#7 SIN `Functions`

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #6 first): SIN^{-1} — Inverse sine sin^{-1}

3rd / g layer (press #1 first): CONST — CONST — scientific-constants menu

Keyboard: `S`

#8 COS `Functions`

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #6 first): COS^{-1} — Inverse cosine cos^{-1}

3rd / g layer (press #1 first): FIX — FIX — fixed decimals

Keyboard: `C`

#9 TAN Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #6 first): TAN^{-1} — Inverse tangent $\tan^{-1}$

3rd / g layer (press #1 first): π — π (3.14159265…) as an exact constant

Keyboard: t

#10 CE/C Clear, delete & power

Primary: CE/C — clear entry, press twice to clear the calculation

Keyboard: Escape

#11 Σ+ Memory, variables & registers

Primary: Σ+ — enter a statistics data point

shifted (2nd / f / SHIFT layer) (press #6 first): Σ− — Σ− — remove a statistics data point

3rd / g layer (press #1 first): STAT1 — STAT1 — one-variable statistics

#12 1/x Functions

Primary: Reciprocal x^{-1} (1/x)

shifted (2nd / f / SHIFT layer) (press #6 first): FRC — Fraction template ▭/▭ (numerator over denominator)

3rd / g layer (press #1 first): STAT2 — STAT2 — two-variable statistics

Keyboard: i

#13 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #6 first): $\sqrt[3]{}$ — Cube root $\sqrt[3]{}$

3rd / g layer (press #1 first): x^3 — Cube x^3

Keyboard: q

#14 √x Functions

Primary: Square root √

shifted (2nd / f / SHIFT layer) (press #6 first): INT — INT — simple interest

3rd / g layer (press #1 first): ABS — Absolute value |x|

Keyboard: r

#15 yx Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #6 first): % — Percent %

3rd / g layer (press #1 first): RND — Round to the displayed precision

Keyboard: ^

#16 π Functions

Primary: π (3.14159265…) as an exact constant

shifted (2nd / f / SHIFT layer) (press #6 first): $\bar{x}$ — $\bar{x}$ — mean

3rd / g layer (press #1 first): $\bar{y}$ — $\bar{y}$ — mean of y

Keyboard: p

#17 EE Digits & number entry

Primary: EXP — enter a power-of-ten exponent (scientific notation)

shifted (2nd / f / SHIFT layer) (press #6 first): DEC — Convert/display in base 10

3rd / g layer (press #1 first): ENG — ENG display mode

#18 (Operators & execute

Primary: Opening bracket (

shifted (2nd / f / SHIFT layer) (press #6 first): HEX — Convert/display in base 16

3rd / g layer (press #1 first): SCI — SCI display mode

Keyboard: (

#19) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

shifted (2nd / f / SHIFT layer) (press #6 first): OCT — Convert/display in base 8

3rd / g layer (press #1 first): FLO — FLO — floating display

Keyboard:)

#20 ÷ Operators & execute

Primary: Division ÷

shifted (2nd / f / SHIFT layer) (press #6 first): BIN — Convert/display in base 2

Keyboard: /

#21 K Functions

Primary: K — constant: repeat the last operation with a fixed operand

shifted (2nd / f / SHIFT layer) (press #6 first): AND — AND (bitwise)

3rd / g layer (press #1 first): A — Hex digit A

#22 7 Digits & number entry

Primary: Digit 7

shifted (2nd / f / SHIFT layer) (press #6 first): OR — OR (bitwise)

3rd / g layer (press #1 first): B — Hex digit B

Keyboard: 7

#23 8 Digits & number entry

Primary: Digit 8

shifted (2nd / f / SHIFT layer) (press #6 first): XOR — XOR (bitwise)

3rd / g layer (press #1 first): C — Hex digit C

Keyboard: 8

#24 9 Digits & number entry

Primary: Digit 9

shifted (2nd / f / SHIFT layer) (press #6 first): XNOR — XNOR (bitwise)

3rd / g layer (press #1 first): D — Hex digit D

Keyboard: 9

#25 × Operators & execute

Primary: Multiplication ×

shifted (2nd / f / SHIFT layer) (press #6 first): NOT — NOT (bitwise)

Keyboard: * x

#26 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

shifted (2nd / f / SHIFT layer) (press #6 first): EXC — EXC — exchange display with a memory

3rd / g layer (press #1 first): E — Hex digit E

#27 4 Digits & number entry

Primary: Digit 4

shifted (2nd / f / SHIFT layer) (press #6 first): nPr — Permutations nPr

3rd / g layer (press #1 first): F — Hex digit F

Keyboard: 4

#28 5 Digits & number entry

Primary: Digit 5

shifted (2nd / f / SHIFT layer) (press #6 first): Sx — Sx — sample standard deviation

3rd / g layer (press #1 first): σx — σ — population standard deviation

Keyboard: 5

#29 6 Digits & number entry

Primary: Digit 6

shifted (2nd / f / SHIFT layer) (press #6 first): Σx — Σx — sum of x

3rd / g layer (press #1 first): Σx^2 — Σx^2 — sum of squares

Keyboard: 6

#30 – Operators & execute

Primary: Subtraction –

shifted (2nd / f / SHIFT layer) (press #6 first): n — n — count of data

Keyboard: -

#31 STO Memory, variables & registers

Primary: STO — store the value into a memory / variable (press the target next)

shifted (2nd / f / SHIFT layer) (press #6 first): F ↔ D — F ↔ D — fraction ⇔ decimal

3rd / g layer (press #1 first): a — a — regression intercept

#32 1 Digits & number entry

Primary: Digit 1

shifted (2nd / f / SHIFT layer) (press #6 first): ► cm — Unit conversion: multiply by 2.54

3rd / g layer (press #1 first): ► in — Unit conversion: multiply by 0.3937007874

Keyboard: 1

#33 2 Digits & number entry

Primary: Digit 2

shifted (2nd / f / SHIFT layer) (press #6 first): ▶ L — Unit conversion: multiply by 3.785411784

3rd / g layer (press #1 first): ▶ gal — Unit conversion: multiply by 0.2641720524

Keyboard: 2

#34 3 Digits & number entry

Primary: Digit 3

shifted (2nd / f / SHIFT layer) (press #6 first): ▶ kg — Unit conversion: multiply by 0.45359237

3rd / g layer (press #1 first): ▶ lb — Unit conversion: multiply by 2.2046226218

Keyboard: 3

#35 + Operators & execute

Primary: Addition +

shifted (2nd / f / SHIFT layer) (press #6 first): b — b — regression slope

3rd / g layer (press #1 first): r — r — correlation coefficient

Keyboard: +

#36 x⇄y Operators & execute

Primary: x≷y — exchange X and Y

shifted (2nd / f / SHIFT layer) (press #6 first): nCr — Combinations nCr

#37 0 Digits & number entry

Primary: Digit 0

shifted (2nd / f / SHIFT layer) (press #6 first): ▶ °C — °F → °C

3rd / g layer (press #1 first): ▶ °F — °C → °F

Keyboard: 0

#38 . Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

shifted (2nd / f / SHIFT layer) (press #6 first): ▶ DMS — Convert decimal degrees to D°M′ S″

3rd / g layer (press #1 first): ▶ DD — Convert D°M′ S″ to decimal degrees

Keyboard: . ,

#39 +/− Digits & number entry

Primary: Change sign (+/−) of the number being typed or the current result

shifted (2nd / f / SHIFT layer) (press #6 first): R▶P — R → P — rectangular to polar

3rd / g layer (press #1 first): P▶R — P → R — polar to rectangular

#40 = Operators & execute

Primary: Equals — complete the calculation

Keyboard: =

5 · Worked examples

23 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

`#32` 1 → `#33` 2 → `#35` + → `#34` 3 → `#25` × → `#27` 4 → `#40` =

Display shows `24.` **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

`#18` (→ `#32` 1 → `#33` 2 → `#35` + → `#23` 8 → `#19`) → `#20` ÷ → `#27` 4 → `#40` =

Display shows `5.` **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

#18 (→ #34 3 → #35 + → #27 4 → #19) → #25 × → #18 (→ #24 9 → #30 − → #33 2 → #19) → #40 =

Display shows 49. **— expected 49. ✔ verified on the app.**

Both brackets are evaluated first, then multiplied.

Example 5.4 — Squares

Calculate 12^2. Press Reset first.

#32 1 → #33 2 → #13 x^2 → #40 =

Display shows 144. **— expected 144. ✔ verified on the app.**

x^2 squares the value — quicker and safer than typing the number twice.

Example 5.5 — Square roots

Calculate $\sqrt{144}$. Press Reset first.

#32 1 → #27 4 → #27 4 → #14 $\sqrt{x}$ → #40 =

Display shows 12. **— expected 12. ✔ verified on the app.**

The square-root key returns the principal (positive) root.

Example 5.6 — Powers

Calculate 2^10. Press Reset first.

#33 2 → #15 y^x → #32 1 → #37 0 → #40 =

Display shows 1024. **— expected 1024. ✔ verified on the app.**

The power key raises the first number to the second.

Example 5.7 — Reciprocals

Calculate 1 ÷ 4 (x^{-1}). Press Reset first.

#27 4 → #12 1/*x* → #40 =

Display shows 0.25 **— expected 0.25. ✔ verified on the app.**

x^{-1} (1/x) gives the reciprocal — handy for resistors in parallel or rates.

Example 5.8 — Trigonometry (degrees)

Calculate tan 45° (Deg mode). Press Reset first.

#27 4 → #28 5 → #9 TAN → #40 =

Display shows 1. **— expected 1. ✔ verified on the app.**

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.9 — Common logarithm

Calculate **log 1000**. Press **Reset** first.

#32 1 → #37 0 → #37 0 → #37 0 → #2 LOG → #40 =

Display shows 3. — expected 3. ✔ verified on the app.

log is base-10: it counts the zeros of a power of ten.

Example 5.10 — Natural logarithm

Calculate **ln 1**. Press **Reset** first.

#32 1 → #3 LN → #40 =

Display shows 0. — expected 0. ✔ verified on the app.

ln is the natural (base-e) logarithm.

Example 5.11 — Antilog 10^x

Calculate **10^2**. Press **Reset** first.

#33 2 → #6 2nd → #2 10^x → #40 =

Display shows 100. — expected 100. ✔ verified on the app.

10^x is the inverse of log.

Example 5.12 — Exponential e^x

Calculate e^1. Press Reset first.

#32 1 → #6 2nd → #3 e^x → #40 =

Display shows 2.718281828 — expected 2.71828183. ✔ verified on the app.

e^x is the inverse of ln.

Example 5.13 — Negative numbers

Calculate −5 + 12. Press Reset first.

#28 5 → #39 +/− → #35 + → #32 1 → #33 2 → #40 =

Display shows 7. — expected 7. ✔ verified on the app.

Use the sign key for a negative number — not the subtraction key.

Example 5.14 — Scientific notation

Calculate 6×10^2 × 3. Press Reset first.

#29 6 → #17 EE → #33 2 → #25 × → #34 3 → #40 =

Display shows 1800. — expected 1800. ✔ verified on the app.

The exponent key enters ×10^n without typing × 10 ^.

Example 5.15 — Area of a circle πr^2

Calculate $\pi \times 3^2$. Press Reset first.

#1 3rd → #9 π → #25 × → #34 3 → #13 x^2 → #40 =

Display shows 28.27433388 **— expected 28.27433388. ✔ verified on the app.**

π is stored to full precision — never type 3.14.

Example 5.16 — Combinations nCr

Calculate 5C2. Press Reset first.

#28 5 → #6 2nd → #36 *nCr* → #33 2 → #40 =

Display shows 10. **— expected 10. ✔ verified on the app.**

nCr counts selections where order does not matter.

Example 5.17 — Factorials

Calculate 5!. Press Reset first.

#28 5 → #1 3rd → #3 *x!* → #40 =

Display shows 120. **— expected 120. ✔ verified on the app.**

n! multiplies 1×2×…×n; 0! is 1 by definition.

Example 5.18 — Continue from the answer

Calculate 100 × 4 = , then Ans ÷ 2. Press Reset first.

#32 1 → #37 0 → #37 0 → #25 × → #27 4 → #40 = → #20 ÷ → #33 2 → #40 =

Display shows 200. **— expected 200. ✔ verified on the app.**

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.19 — Percent of an amount

Calculate 15 % of 80. Press Reset first.

#23 8 → #37 0 → #25 × → #32 1 → #28 5 → #6 2nd → #15 % → #40 =

Display shows 12. **— expected 12. ✔ verified on the app.**

The percent key turns the second number into a percentage of the first.

Example 5.20 — Average of three numbers

Calculate (4 + 8 + 15) ÷ 3. Press Reset first.

#18 (→ #27 4 → #35 + → #23 8 → #35 + → #32 1 → #28 5 → #19) → #20 ÷ → #34 3 → #40 =

Display shows 9. **— expected 9. ✔ verified on the app.**

Add up the values, then divide by how many there are.

Example 5.21 — Mixed operations

Calculate 50 − 4 × 6 + 7. Press Reset first.

#28 5 → #37 0 → #30 − → #27 4 → #25 × → #29 6 → #35 + → #22 7 → #40 =

Display shows 33. **— expected 33. ✔ verified on the app.**

Only the × part is done first; the − and + are then worked left to right.

Example 5.22 — Dividing twice

Calculate 360 ÷ 4 ÷ 9. Press Reset first.

#34 3 → #29 6 → #37 0 → #20 ÷ → #27 4 → #20 ÷ → #24 9 → #40 =

Display shows 10. **— expected 10. ✔ verified on the app.**

Division is worked left to right: the first quotient is divided again.

Example 5.23 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#18 (→ #18 (→ #34 3 → #35 + → #28 5 → #19) → #25 × → #27 4 → #30 − → #33 2 → #19) → #20 ÷ → #32 1 → #37 0 → #40 =

Display shows 3. **— expected 3. ✔ verified on the app.**

The innermost brackets are worked out first, then the outer brackets, then the division.

6 · Tricks & hacks

Hack 1: Order of operations

Algebraic one-line models respect × ÷ before + −, so 2 + 3 × 4 = 14 without brackets.

Hack 2: Constant calculations

Repeat the last operation by pressing = again (where supported) — handy for repeated discounts or growth.

Hack 3: Memory

Use the memory keys to hold a subtotal while you calculate something else.

Hack 4: Sign change

Use +/− after typing the number to make it negative.

Hack 5: Statistics

Enter each data value with Σ+/DATA; then read mean and deviation with the shifted keys.

Hack 6: Hidden functions on this model

66 extra functions sit on shifted layers. Press the layer key once, then the key: for example #6 then #2 → 10x; #1 then #2 → DRG▶; #6 then #3 → ex; #1 then #3 → x!.

Hack 7: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #40.

Hack 8: Type faster on a computer

29 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 9: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 10: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 11: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
l	#2	LOG	Common logarithm $\log_{10}$
n	#3	LN	Natural logarithm ln
s	#7	SIN	Sine (uses the current angle unit)
c	#8	COS	Cosine (current angle unit)
t	#9	TAN	Tangent (current angle unit)
Escape	#10	CE/C	CE/C — clear entry, press twice to clear the calculation
i	#12	1/x	Reciprocal x^{-1} (1/x)
q	#13	x^2	Square x^2
r	#14	√x	Square root √
^	#15	yx	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
p	#16	π	π (3.14159265…) as an exact constant
(	#18	(	Opening bracket (
)	#19	)	Closing bracket). Closing brackets at the end of a line may be omitted
/	#20	÷	Division ÷
7	#22	7	Digit 7
8	#23	8	Digit 8
9	#24	9	Digit 9
* x	#25	×	Multiplication ×
4	#27	4	Digit 4
5	#28	5	Digit 5
6	#29	6	Digit 6
-	#30	−	Subtraction −

1	#32	1	Digit 1
2	#33	2	Digit 2
3	#34	3	Digit 3
+	#35	+	Addition +
0	#37	0	Digit 0
. ,	#38	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
=	#40	=	Equals — complete the calculation

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Typing +/− first

On one-line models +/− is pressed after the digits: 5 +/− gives −5.

Mistake 2: CE vs C

CE clears only the last entry; C/AC clears the whole calculation.

Mistake 3: Forgetting the angle unit

Check DEG/RAD/GRAD before trig functions.

Mistake 4: Functions after the number

One-line models apply functions to the number already on display: 9 √ = 3 (not √ 9).

Mistake 5: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 6: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (124)

Work each one on calculator #12, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 1000 + 25 × 4 = _____
7. 19 + 21 × 3 = _____
8. 64 + 8 × 8 = _____
9. (45 + 15) ÷ 6 = _____
10. (9 + 7) ÷ 2 = _____
11. (100 + 60) ÷ 8 = _____
12. (33 + 17) ÷ 5 = _____
13. (1 + 2) ÷ 4 = _____
14. (250 + 350) ÷ 12 = _____
15. (5 + 6) × (10 − 3) = _____
16. (12 + 3) × (8 − 5) = _____

17. (1.5 + 2.5) × (7 − 4) = _____

18. (20 + 30) × (15 − 5) = _____

19. (9 + 1) × (100 − 1) = _____

20. (11 + 13) × (17 − 9) = _____

21. 25^2 = _____

22. 7^2 = _____

23. 1.5^2 = _____

24. 19^2 = _____

25. 33^2 = _____

26. 0.5^2 = _____

27. 125^2 = _____

28. 64^2 = _____

29. $\sqrt{625}$ = _____

30. $\sqrt{2}$ = _____

31. $\sqrt{50}$ = _____

32. $\sqrt{1024}$ = _____

33. $\sqrt{3}$ = _____

34. $\sqrt{9801}$ = _____

35. 3^5 = _____

36. 1.05^3 = _____

37. 5^4 = _____

38. 7^3 = _____

39. 1.5^2 = _____

40. 10^6 = _____

41. 2^0.5 = _____

42. 1 ÷ 8 (x^{-1}) = _____

43. 1 ÷ 0.25 (x^{-1}) = _____

44. 1 ÷ 16 (x^{-1}) = _____

45. 1 ÷ 40 (x^{-1}) = _____

46. 1 ÷ 0.2 (x^{-1}) = _____

47. sin 90° (Deg mode) = _____

48. cos 0° (Deg mode) = _____

49. log 100 = _____

50. log 10000 = _____

51. ln 10 = _____

52. ln 2 = _____

53. 10^3 = _____

54. e^2 = _____

55. −20 + 7 = _____

56. −3 + 3 = _____

57. −100 + 250 = _____

58. −8 + 1 = _____

59. 2×10^3 × 4 = _____

60. 1.5×10^4 × 2 = _____

61. 3×10^5 × 3 = _____

62. 4.5×10^2 × 2 = _____

63. π × 10^2 = _____

64. π × 2.5^2 = _____

65. π × 7^2 = _____

66. π × 1.2^2 = _____

67. 10C3 = _____

68. 6C6 = _____

69. 6! = _____

70. 0! = _____

71. 15 × 6 = , then Ans ÷ 9 = _____

72. 7.5 × 8 = , then Ans ÷ 3 = _____

73. 250 × 12 = , then Ans ÷ 100 = _____

74. 8 % of 250 = _____

75. 25 % of 360 = _____

76. 12.5 % of 40 = _____

77. 10 % of 450 = _____

78. 20 % of 125 = _____

79. 5 % of 640 = _____

80. 75 % of 48 = _____

81. 30 % of 90 = _____

82. (12 + 15 + 18) ÷ 3 = _____

83. (70 + 85 + 91) ÷ 3 = _____

84. (2.5 + 3.5 + 6) ÷ 3 = _____

85. (100 + 200 + 600) ÷ 3 = _____

86. 100 − 3 × 12 + 9 = _____

87. 18 − 2 × 5 + 40 = _____

88. 7.5 − 1.5 × 3 + 2 = _____

89. 64 − 8 × 4 + 10 = _____

90. 300 − 25 × 4 + 55 = _____

91. 9 − 3 × 3 + 1 = _____

92. 45 − 5 × 5 + 15 = _____

93. 1000 ÷ 8 ÷ 5 = _____

94. 96 ÷ 2 ÷ 3 = _____

95. 7200 ÷ 60 ÷ 60 = _____

96. 500 ÷ 4 ÷ 25 = _____

97. 81 ÷ 3 ÷ 3 = _____

98. ((10 + 2) × 3 − 6) ÷ 6 = _____

99. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

100. ((20 + 30) × 2 − 25) ÷ 3 = _____

101. ((7 + 8) × 9 − 15) ÷ 12 = _____

102. 12 + 3 × 4 = _____

103. (12 + 8) ÷ 4 = _____

104. (3 + 4) × (9 − 2) = _____

105. 12^2 = _____

106. $\sqrt{144}$ = _____

107. 2^10 = _____

108. 1 ÷ 4 (x^{-1}) = _____

109. tan 45° (Deg mode) = _____

110. log 1000 = _____

111. ln 1 = _____

112. 10^2 = _____

113. e^1 = _____

114. −5 + 12 = _____

115. 6×10^2 × 3 = _____

116. π × 3^2 = _____

117. 5C2 = _____

118. 5! = _____

119. 100 × 4 = , then Ans ÷ 2 = _____

120. 15 % of 80 = _____

121. (4 + 8 + 15) ÷ 3 = _____

122. 50 − 4 × 6 + 7 = _____

123. 360 ÷ 4 ÷ 9 = _____

124. ((3 + 5) × 4 − 2) ÷ 10 = _____

Set B — Key-finding drills (96)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “3rd” (3rd)? _____
2. Which key is “LOG” (Common logarithm $\log_{10}$)? _____
3. Which key is “LN” (Natural logarithm ln)? _____
4. Which key is “DRG” (DRG)? _____
5. Which key is “ON/AC” (ON / AC)? _____
6. Which key is “2nd” (SHIFT / 2nd)? _____
7. Which key is “SIN” (Sine)? _____
8. Which key is “COS” (Cosine)? _____
9. Which key is “TAN” (Tangent)? _____
10. Which key is “CE/C” (CE/C)? _____

11. Which key is "Σ+" (Σ+)? _____

12. Which key is "1/x" (Reciprocal x^{-1})? _____

13. Which key is "x^2" (Square x^2)? _____

14. Which key is "√x" (Square root √)? _____

15. Which key is "yx" (Power x^y / ^)? _____

16. Which key is "π" (π)? _____

17. Which key is "EE" (EXP)? _____

18. Which key is "(" (Opening bracket)? _____

19. Which key is ")" (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

20. Which key is "÷" (Division ÷)? _____

21. Which key is "K" (K)? _____

22. Which key is "×" (Multiplication ×)? _____

23. Which key is "RCL" (RCL)? _____

24. Which key is "−" (Subtraction −)? _____

25. Which key is "STO" (STO)? _____

26. Which key is "+" (Addition +)? _____

27. Which key is "x⇄y" (x≷y)? _____

28. Which key is "." (Decimal point. Pressing it first gives "0."; a second point in the same number is ignored)? _____

29. Which key is "+/−" (Change sign)? _____

30. Which key is "=" (Equals)? _____

31. How do you reach "10x"? _____

32. How do you reach "DRG▶"? _____

33. How do you reach "ex"? _____

34. How do you reach "x!"? _____

35. How do you reach "$x\sqrt{\ }y$"? _____

36. How do you reach "HYP"? _____

37. How do you reach "SIN^{-1}"? _____

38. How do you reach "CONST"? _____

39. How do you reach "COS^{-1}"? _____

40. How do you reach "FIX"? _____

41. How do you reach "TAN^{-1}"? _____

42. How do you reach "π"? _____

43. How do you reach "Σ–"? _____

44. How do you reach "STAT1"? _____

45. How do you reach "FRC"? _____

46. How do you reach "STAT2"? _____

47. How do you reach "$\sqrt[3]{\ }$"? _____

48. How do you reach "x^3"? _____

49. How do you reach "INT"? _____

50. How do you reach “ABS”? _____

51. How do you reach “%”? _____

52. How do you reach “RND”? _____

53. How do you reach “$\bar{x}$”? _____

54. How do you reach “$\bar{y}$”? _____

55. How do you reach “DEC”? _____

56. How do you reach “ENG”? _____

57. How do you reach “HEX”? _____

58. How do you reach “SCI”? _____

59. How do you reach “OCT”? _____

60. How do you reach “FLO”? _____

61. How do you reach “BIN”? _____

62. How do you reach “AND”? _____

63. How do you reach “A”? _____

64. How do you reach “OR”? _____

65. How do you reach “B”? _____

66. How do you reach “XOR”? _____

67. How do you reach “C”? _____

68. How do you reach “XNOR”? _____

69. How do you reach “D”? _____

70. How do you reach “NOT”? _____

71. How do you reach “EXC”? _____

72. How do you reach “E”? _____

73. How do you reach “nPr”? _____

74. How do you reach “F”? _____

75. How do you reach “Sx”? _____

76. How do you reach “σx”? _____

77. How do you reach “Σx”? _____

78. How do you reach “Σx^2”? _____

79. How do you reach “n”? _____

80. How do you reach “F ↔ D”? _____

81. How do you reach “a”? _____

82. How do you reach “ ► cm”? _____

83. How do you reach “ ► in”? _____

84. How do you reach “ ► L”? _____

85. How do you reach “ ► gal”? _____

86. How do you reach “ ► kg”? _____

87. How do you reach “ ► lb”? _____

88. How do you reach “b”? _____

89. How do you reach “r”? _____

90. How do you reach “nCr”? _____

91. How do you reach “►°C”? _____

92. How do you reach “►°F”? _____

93. How do you reach “►DMS”? _____

94. How do you reach “►DD”? _____

95. How do you reach “R▶P”? _____

96. How do you reach “P▶R”? _____

Appendix A — Drill answers

Set A

1. 55 (display `55.`)

`#33` 2 → `#28` 5 → `#35` + → `#28` 5 → `#25` × → `#29` 6 → `#40` =

2. 79 (display `79.`)

`#22` 7 → `#35` + → `#23` 8 → `#25` × → `#24` 9 → `#40` =

3. 282 (display `282.`)

`#32` 1 → `#28` 5 → `#37` 0 → `#35` + → `#32` 1 → `#33` 2 → `#25` × → `#32` 1 → `#32` 1 → `#40` =

4. 213 (display `213.`)

`#34` 3 → `#35` + → `#32` 1 → `#27` 4 → `#25` × → `#32` 1 → `#28` 5 → `#40` =

5. 90 (display `90.`)

`#27` 4 → `#23` 8 → `#35` + → `#29` 6 → `#25` × → `#22` 7 → `#40` =

6. 1100 (display `1100.`)

`#32` 1 → `#37` 0 → `#37` 0 → `#37` 0 → `#35` + → `#33` 2 → `#28` 5 → `#25` × → `#27` 4 → `#40` =

7. 82 (display `82.`)

`#32` 1 → `#24` 9 → `#35` + → `#33` 2 → `#32` 1 → `#25` × → `#34` 3 → `#40` =

8. 128 (display `128.`)

`#29` 6 → `#27` 4 → `#35` + → `#23` 8 → `#25` × → `#23` 8 → `#40` =

9. 10 (display `10.`)

`#18` (→ `#27` 4 → `#28` 5 → `#35` + → `#32` 1 → `#28` 5 → `#19`) → `#20` ÷ → `#29` 6 → `#40` =

10. 8 (display `8.`)

`#18` (→ `#24` 9 → `#35` + → `#22` 7 → `#19`) → `#20` ÷ → `#33` 2 → `#40` =

11. 20 (display `20.`)

`#18` (→ `#32` 1 → `#37` 0 → `#37` 0 → `#35` + → `#29` 6 → `#37` 0 → `#19`) → `#20` ÷ → `#23` 8 → `#40` =

12. 10 (display `10.`)

`#18` (→ `#34` 3 → `#34` 3 → `#35` + → `#32` 1 → `#22` 7 → `#19`) → `#20` ÷ → `#28` 5 → `#40` =

13. 0.75 (display `0.75`)

`#18` (→ `#32` 1 → `#35` + → `#33` 2 → `#19`) → `#20` ÷ → `#27` 4 → `#40` =

14. 50 (display 50.)

#18 (→ #33 2 → #28 5 → #37 0 → #35 + → #34 3 → #28 5 → #37 0 → #19) → #20 ÷ → #32 1 → #33 2 → #40 =

15. 77 (display 77.)

#18 (→ #28 5 → #35 + → #29 6 → #19) → #25 × → #18 (→ #32 1 → #37 0 → #30 − → #34 3 → #19) → #40 =

16. 45 (display 45.)

#18 (→ #32 1 → #33 2 → #35 + → #34 3 → #19) → #25 × → #18 (→ #23 8 → #30 − → #28 5 → #19) → #40 =

17. 12 (display 12.)

#18 (→ #32 1 → #38 . → #28 5 → #35 + → #33 2 → #38 . → #28 5 → #19) → #25 × → #18 (→ #22 7 → #30 − → #27 4 → #19) → #40 =

18. 500 (display 500.)

#18 (→ #33 2 → #37 0 → #35 + → #34 3 → #37 0 → #19) → #25 × → #18 (→ #32 1 → #28 5 → #30 − → #28 5 → #19) → #40 =

19. 990 (display 990.)

#18 (→ #24 9 → #35 + → #32 1 → #19) → #25 × →
#18 (→ #32 1 → #37 0 → #37 0 → #30 − → #32 1 →
#19) → #40 =

20. 192 (display 192.)

#18 (→ #32 1 → #32 1 → #35 + → #32 1 → #34 3 →
#19) → #25 × → #18 (→ #32 1 → #22 7 → #30 − →
#24 9 → #19) → #40 =

21. 625 (display 625.)

#33 2 → #28 5 → #13 x^2 → #40 =

22. 49 (display 49.)

#22 7 → #13 x^2 → #40 =

23. 2.25 (display 2.25)

#32 1 → #38 . → #28 5 → #13 x^2 → #40 =

24. 361 (display 361.)

#32 1 → #24 9 → #13 x^2 → #40 =

25. 1089 (display 1089.)

#34 3 → #34 3 → #13 x^2 → #40 =

26. 0.25 (display 0.25)

#37 0 → #38 . → #28 5 → #13 x^2 → #40 =

27. 15625 (display `15625.`)

`#32` 1 → `#33` 2 → `#28` 5 → `#13` x^2 → `#40` =

28. 4096 (display `4096.`)

`#29` 6 → `#27` 4 → `#13` x^2 → `#40` =

29. 25 (display `25.`)

`#29` 6 → `#33` 2 → `#28` 5 → `#14` $\sqrt{x}$ → `#40` =

30. 1.41421356 (display `1.414213562`)

`#33` 2 → `#14` $\sqrt{x}$ → `#40` =

31. 7.07106781 (display `7.071067812`)

`#28` 5 → `#37` 0 → `#14` $\sqrt{x}$ → `#40` =

32. 32 (display `32.`)

`#32` 1 → `#37` 0 → `#33` 2 → `#27` 4 → `#14` $\sqrt{x}$ → `#40` =

33. 1.73205081 (display `1.732050808`)

`#34` 3 → `#14` $\sqrt{x}$ → `#40` =

34. 99 (display `99.`)

`#24` 9 → `#23` 8 → `#37` 0 → `#32` 1 → `#14` $\sqrt{x}$ → `#40` =

35. 243 (display `243.`)

`#34` 3 → `#15` y^x → `#28` 5 → `#40` =

36. 1.157625 (display `1.157625`)

`#32` 1 → `#38` . → `#37` 0 → `#28` 5 → `#15` y^x → `#34` 3 → `#40` =

37. 625 (display `625.`)

`#28` 5 → `#15` y^x → `#27` 4 → `#40` =

38. 343 (display `343.`)

`#22` 7 → `#15` y^x → `#34` 3 → `#40` =

39. 2.25 (display `2.25`)

`#32` 1 → `#38` . → `#28` 5 → `#15` y^x → `#33` 2 → `#40` =

40. 1000000.000 (display `1000000.`)

`#32` 1 → `#37` 0 → `#15` y^x → `#29` 6 → `#40` =

41. 1.41421356 (display `1.414213562`)

`#33` 2 → `#15` y^x → `#37` 0 → `#38` . → `#28` 5 → `#40` =

42. 0.125 (display `0.125`)

`#23` 8 → `#12` $1/x$ → `#40` =

43. 4 (display `4.`)

`#37` 0 → `#38` . → `#33` 2 → `#28` 5 → `#12` $1/x$ → `#40` =

44. 0.0625 (display `0.0625`)

`#32` 1 → `#29` 6 → `#12` $1/x$ → `#40` =

45. 0.025 (display `0.025`)

`#27` 4 → `#37` 0 → `#12` 1/*x* → `#40` =

46. 5 (display `5.`)

`#37` 0 → `#38` . → `#33` 2 → `#12` 1/*x* → `#40` =

47. 1 (display `1.`)

`#24` 9 → `#37` 0 → `#7` SIN → `#40` =

48. 1 (display `1.`)

`#37` 0 → `#8` COS → `#40` =

49. 2 (display `2.`)

`#32` 1 → `#37` 0 → `#37` 0 → `#2` LOG → `#40` =

50. 4 (display `4.`)

`#32` 1 → `#37` 0 → `#37` 0 → `#37` 0 → `#37` 0 → `#2` LOG → `#40` =

51. 2.30258509 (display `2.302585093`)

`#32` 1 → `#37` 0 → `#3` LN → `#40` =

52. 0.69314718 (display `0.6931471806`)

`#33` 2 → `#3` LN → `#40` =

53. 1000 (display `1000.`)

`#34` 3 → `#6` 2nd → `#2` 10^x → `#40` =

54. 7.3890561 (display `7.389056099`)

`#33` 2 → `#6` 2nd → `#3` e^x → `#40` =

55. -13 (display `-13.`)

`#33` 2 → `#37` 0 → `#39` +/− → `#35` + → `#22` 7 → `#40` =

56. 0 (display `0.`)

`#34` 3 → `#39` +/− → `#35` + → `#34` 3 → `#40` =

57. 150 (display `150.`)

`#32` 1 → `#37` 0 → `#37` 0 → `#39` +/− → `#35` + → `#33` 2 → `#28` 5 → `#37` 0 → `#40` =

58. -7 (display `-7.`)

`#23` 8 → `#39` +/− → `#35` + → `#32` 1 → `#40` =

59. 8000 (display `8000.`)

`#33` 2 → `#17` EE → `#34` 3 → `#25` × → `#27` 4 → `#40` =

60. 30000 (display `30000.`)

`#32` 1 → `#38` . → `#28` 5 → `#17` EE → `#27` 4 → `#25` × → `#33` 2 → `#40` =

61. 900000 (display `900000.`)

`#34` 3 → `#17` EE → `#28` 5 → `#25` × → `#34` 3 → `#40` =

62. 900 (display `900.`)

`#27` 4 → `#38` . → `#28` 5 → `#17` EE → `#33` 2 → `#25` × → `#33` 2 → `#40` =

63. 314.15926536 (display 314.1592654)

#1 3rd → #9 π → #25 × → #32 1 → #37 0 → #13 x^2 → #40 =

64. 19.63495408 (display 19.63495408)

#1 3rd → #9 π → #25 × → #33 2 → #38 . → #28 5 → #13 x^2 → #40 =

65. 153.93804003 (display 153.93804)

#1 3rd → #9 π → #25 × → #22 7 → #13 x^2 → #40 =

66. 4.52389342 (display 4.523893421)

#1 3rd → #9 π → #25 × → #32 1 → #38 . → #33 2 → #13 x^2 → #40 =

67. 120 (display 120.)

#32 1 → #37 0 → #6 2nd → #36 *n*C*r* → #34 3 → #40 =

68. 1 (display 1.)

#29 6 → #6 2nd → #36 *n*C*r* → #29 6 → #40 =

69. 720 (display 720.)

#29 6 → #1 3rd → #3 *x*! → #40 =

70. 1 (display 1.)

#37 0 → #1 3rd → #3 *x*! → #40 =

71. 10 (display `10.`)

`#32` 1 → `#28` 5 → `#25` × → `#29` 6 → `#40` = → `#20` ÷ → `#24` 9 → `#40` =

72. 20 (display `20.`)

`#22` 7 → `#38` . → `#28` 5 → `#25` × → `#23` 8 → `#40` = → `#20` ÷ → `#34` 3 → `#40` =

73. 30 (display `30.`)

`#33` 2 → `#28` 5 → `#37` 0 → `#25` × → `#32` 1 → `#33` 2 → `#40` = → `#20` ÷ → `#32` 1 → `#37` 0 → `#37` 0 → `#40` =

74. 20 (display `20.`)

`#33` 2 → `#28` 5 → `#37` 0 → `#25` × → `#23` 8 → `#6` 2nd → `#15` % → `#40` =

75. 90 (display `90.`)

`#34` 3 → `#29` 6 → `#37` 0 → `#25` × → `#33` 2 → `#28` 5 → `#6` 2nd → `#15` % → `#40` =

76. 5 (display `5.`)

`#27` 4 → `#37` 0 → `#25` × → `#32` 1 → `#33` 2 → `#38` . → `#28` 5 → `#6` 2nd → `#15` % → `#40` =

77. 45 (display `45.`)

`#27` 4 → `#28` 5 → `#37` 0 → `#25` × → `#32` 1 → `#37` 0 → `#6` 2nd → `#15` % → `#40` =

78. 25 (display `25.`)

`#32` 1 → `#33` 2 → `#28` 5 → `#25` × → `#33` 2 → `#37` 0 → `#6` 2nd → `#15` % → `#40` =

79. 32 (display `32.`)

`#29` 6 → `#27` 4 → `#37` 0 → `#25` × → `#28` 5 → `#6` 2nd → `#15` % → `#40` =

80. 36 (display `36.`)

`#27` 4 → `#23` 8 → `#25` × → `#22` 7 → `#28` 5 → `#6` 2nd → `#15` % → `#40` =

81. 27 (display `27.`)

`#24` 9 → `#37` 0 → `#25` × → `#34` 3 → `#37` 0 → `#6` 2nd → `#15` % → `#40` =

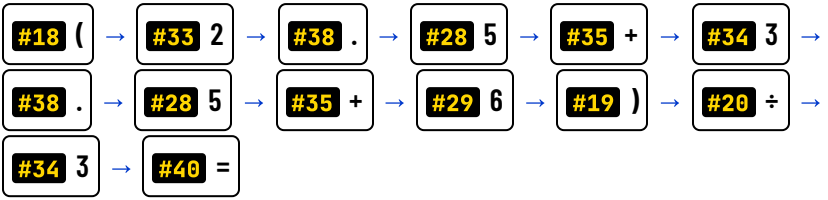
82. 15 (display `15.`)

`#18` (→ `#32` 1 → `#33` 2 → `#35` + → `#32` 1 → `#28` 5 → `#35` + → `#32` 1 → `#23` 8 → `#19`) → `#20` ÷ → `#34` 3 → `#40` =

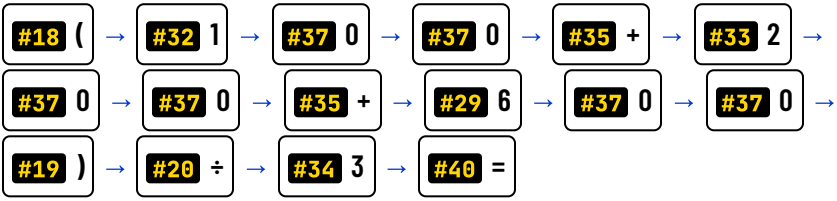
83. 82 (display `82.`)

`#18` (→ `#22` 7 → `#37` 0 → `#35` + → `#23` 8 → `#28` 5 → `#35` + → `#24` 9 → `#32` 1 → `#19`) → `#20` ÷ → `#34` 3 → `#40` =

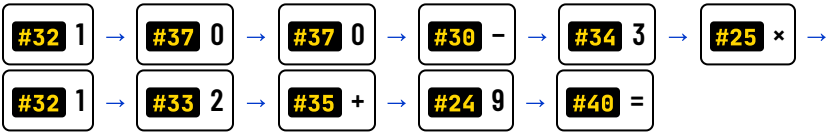
84. 4 (display `4.`)



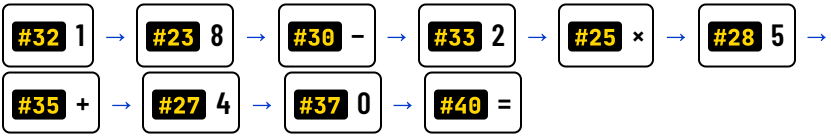
85. 300 (display `300.`)



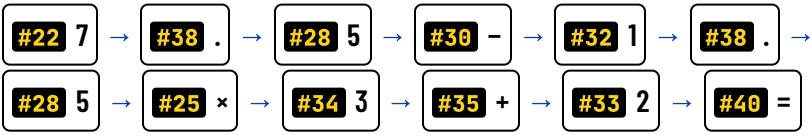
86. 73 (display `73.`)



87. 48 (display `48.`)



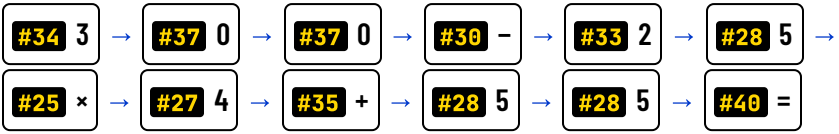
88. 5 (display `5.`)



89. 42 (display `42.`)



90. 255 (display `255.`)



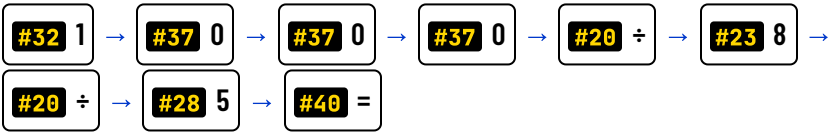
91. 1 (display `1.`)



92. 35 (display `35.`)



93. 25 (display `25.`)



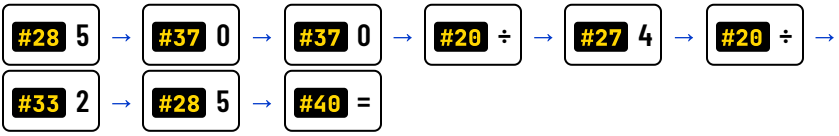
94. 16 (display `16.`)



95. 2 (display `2.`)



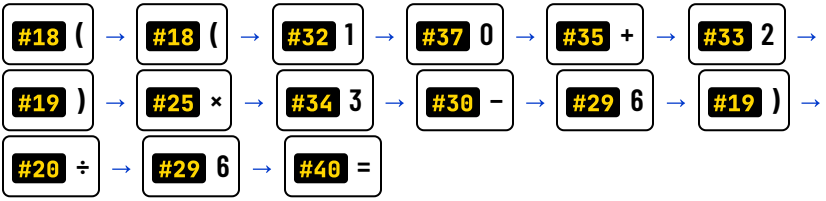
96. 5 (display `5.`)



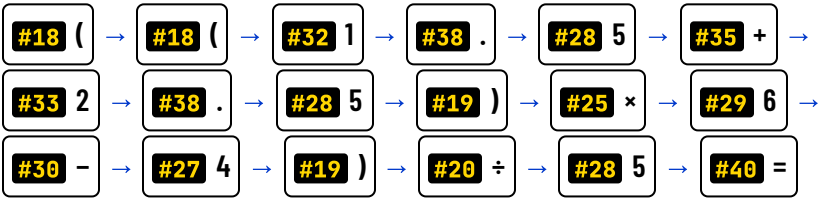
97. 9 (display `9.`)



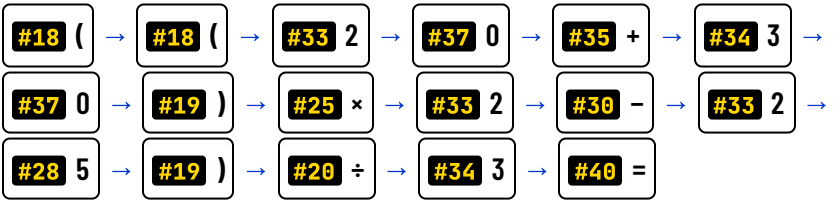
98. 5 (display `5.`)



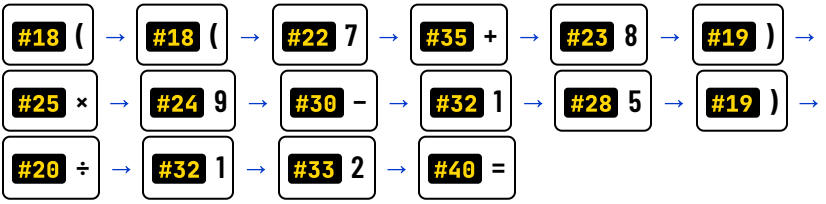
99. 4 (display `4.`)



100. 25 (display `25.`)



101. 10 (display `10.`)



102. 24 (display `24.`)

`#32` 1 → `#33` 2 → `#35` + → `#34` 3 → `#25` × → `#27` 4 → `#40` =

103. 5 (display `5.`)

`#18` (→ `#32` 1 → `#33` 2 → `#35` + → `#23` 8 → `#19`) → `#20` ÷ → `#27` 4 → `#40` =

104. 49 (display `49.`)

`#18` (→ `#34` 3 → `#35` + → `#27` 4 → `#19`) → `#25` × → `#18` (→ `#24` 9 → `#30` − → `#33` 2 → `#19`) → `#40` =

105. 144 (display `144.`)

`#32` 1 → `#33` 2 → `#13` x^2 → `#40` =

106. 12 (display `12.`)

`#32` 1 → `#27` 4 → `#27` 4 → `#14` $\sqrt{x}$ → `#40` =

107. 1024 (display `1024.`)

`#33` 2 → `#15` y^x → `#32` 1 → `#37` 0 → `#40` =

108. 0.25 (display `0.25`)

`#27` 4 → `#12` $1/x$ → `#40` =

109. 1 (display `1.`)

`#27` 4 → `#28` 5 → `#9` TAN → `#40` =

110. 3 (display `3.`)

`#32` 1 → `#37` 0 → `#37` 0 → `#37` 0 → `#2` LOG → `#40` =

111. 0 (display `0.`)

`#32` 1 → `#3` LN → `#40` =

112. 100 (display `100.`)

`#33` 2 → `#6` 2nd → `#2` 10^x → `#40` =

113. 2.71828183 (display `2.718281828`)

`#32` 1 → `#6` 2nd → `#3` e^x → `#40` =

114. 7 (display `7.`)

`#28` 5 → `#39` +/− → `#35` + → `#32` 1 → `#33` 2 → `#40` =

115. 1800 (display `1800.`)

`#29` 6 → `#17` EE → `#33` 2 → `#25` × → `#34` 3 → `#40` =

116. 28.27433388 (display `28.27433388`)

`#1` 3rd → `#9` π → `#25` × → `#34` 3 → `#13` x^2 → `#40` =

117. 10 (display `10.`)

`#28` 5 → `#6` 2nd → `#36` *nCr* → `#33` 2 → `#40` =

118. 120 (display `120.`)

`#28` 5 → `#1` 3rd → `#3` *x!* → `#40` =

119. 200 (display `200.`)

`#32` 1 → `#37` 0 → `#37` 0 → `#25` × → `#27` 4 → `#40` = →
`#20` ÷ → `#33` 2 → `#40` =

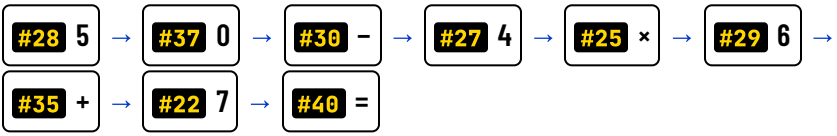
120. 12 (display `12.`)



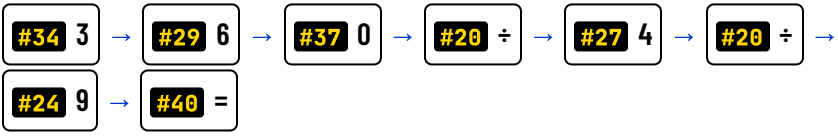
121. 9 (display `9.`)



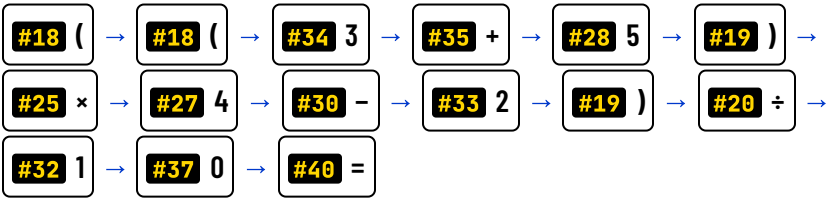
122. 33 (display `33.`)



123. 10 (display `10.`)



124. 3 (display `3.`)



Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14

15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #25
23. #26
24. #30
25. #31
26. #35
27. #36
28. #38
29. #39
30. #40
31. #6 2nd then #2 LOG
32. #1 3rd then #2 LOG
33. #6 2nd then #3 LN
34. #1 3rd then #3 LN
35. #6 2nd then #4 DRG
36. #1 3rd then #4 DRG
37. #6 2nd then #7 SIN
38. #1 3rd then #7 SIN
39. #6 2nd then #8 COS
40. #1 3rd then #8 COS
41. #6 2nd then #9 TAN
42. #1 3rd then #9 TAN
43. #6 2nd then #11 Σ+
44. #1 3rd then #11 Σ+
45. #6 2nd then #12 1/x
46. #1 3rd then #12 1/x
47. #6 2nd then #13 x²
48. #1 3rd then #13 x²
49. #6 2nd then #14 √x
50. #1 3rd then #14 √x
51. #6 2nd then #15 yx
52. #1 3rd then #15 yx
53. #6 2nd then #16 π
54. #1 3rd then #16 π
55. #6 2nd then #17 EE
56. #1 3rd then #17 EE
57. #6 2nd then #18 (
58. #1 3rd then #18 (
59. #6 2nd then #19)
60. #1 3rd then #19)
61. #6 2nd then #20 ÷
62. #6 2nd then #21 K
63. #1 3rd then #21 K
64. #6 2nd then #22 7
65. #1 3rd then #22 7
66. #6 2nd then #23 8
67. #1 3rd then #23 8
68. #6 2nd then #24 9
69. #1 3rd then #24 9
70. #6 2nd then #25 ×
71. #6 2nd then #26 RCL
72. #1 3rd then #26 RCL
73. #6 2nd then #27 4
74. #1 3rd then #27 4
75. #6 2nd then #28 5
76. #1 3rd then #28 5
77. #6 2nd then #29 6
78. #1 3rd then #29 6
79. #6 2nd then #30 -
80. #6 2nd then #31 STO
81. #1 3rd then #31 STO
82. #6 2nd then #32 1
83. #1 3rd then #32 1
84. #6 2nd then #33 2
85. #1 3rd then #33 2
86. #6 2nd then #34 3
87. #1 3rd then #34 3
88. #6 2nd then #35 +
89. #1 3rd then #35 +
90. #6 2nd then #36 x⇄y
91. #6 2nd then #37 0
92. #1 3rd then #37 0
93. #6 2nd then #38 .
94. #1 3rd then #38 .
95. #6 2nd then #39 +/-
96. #1 3rd then #39 +/-

Appendix B — Function finder (A–Z)

Function	Label	Keys
°C → °F	►°F	#1 → #37
°F → °C	►°C	#6 → #37
10^x — antilog	10x	#6 → #2
3rd — the next key uses its third function	3rd	#1
a — regression intercept	a	#1 → #31
Absolute value |x|	ABS	#1 → #14
Addition +	+	#35
AND (bitwise)	AND	#6 → #21
b — regression slope	b	#6 → #35
CE/C — clear entry, press twice to clear the calculation	CE/C	#10
Change sign (+/−) of the number being typed or the current result	+/−	#39
Closing bracket). Closing brackets at the end of a line may be omitted	)	#19
Combinations nCr	nCr	#6 → #36
Common logarithm $\log_{10}$	LOG	#2
CONST — scientific-constants menu	CONST	#1 → #7
Convert D°M′ S″ to decimal degrees	►DD	#1 → #38
Convert decimal degrees to D°M′ S″	►DMS	#6 → #38
Convert/display in base 10	DEC	#6 → #17

Convert/display in base 16	HEX	#6 → #18
Convert/display in base 2	BIN	#6 → #20
Convert/display in base 8	OCT	#6 → #19
Cosine (current angle unit)	COS	#8
Cube root $\sqrt[3]{}$	$\sqrt[3]{}$	#6 → #13
Cube x^3	x^3	#1 → #13
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#38
Digit 0	0	#37
Digit 1	1	#32
Digit 2	2	#33
Digit 3	3	#34
Digit 4	4	#27
Digit 5	5	#28
Digit 6	6	#29
Digit 7	7	#22
Digit 8	8	#23
Digit 9	9	#24
Division ÷	÷	#20
DRG — choose/convert the angle unit (Deg / Rad / Grad)	DRG	#4
DRG▶ — convert the displayed angle to the next unit	DRG▶	#1 → #2
ENG display mode	ENG	#1 → #17
Equals — complete the calculation	=	#40
e^x — exponential	ex	#6 → #3
EXC — exchange display with a memory	EXC	#6 → #26

EXP — enter a power-of-ten exponent (scientific notation)	**EE**	**#17**
F ↔ D — fraction ⇔ decimal	**F ↔ D**	**#6 → #31**
Factorial n! (also on non-integers via the gamma function where supported)	**x!**	**#1 → #3**
FIX — fixed decimals	**FIX**	**#1 → #8**
FLO — floating display	**FLO**	**#1 → #19**
Fraction template ▭/▭ (numerator over denominator)	**FRC**	**#6 → #12**
Hex digit A	**A**	**#1 → #21**
Hex digit B	**B**	**#1 → #22**
Hex digit C	**C**	**#1 → #23**
Hex digit D	**D**	**#1 → #24**
Hex digit E	**E**	**#1 → #26**
Hex digit F	**F**	**#1 → #27**
Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh	**HYP**	**#1 → #4**
INT — simple interest	**INT**	**#6 → #14**
Inverse cosine $\cos^{-1}$	**$\cos^{-1}$**	**#6 → #8**
Inverse sine $\sin^{-1}$	**$\sin^{-1}$**	**#6 → #7**
Inverse tangent $\tan^{-1}$	**$\tan^{-1}$**	**#6 → #9**
K — constant: repeat the last operation with a fixed operand	**K**	**#21**
Multiplication ×	**×**	**#25**
n — count of data	**n**	**#6 → #30**

Natural logarithm ln	LN	#3
NOT (bitwise)	NOT	#6 → #25
ON / AC — power on, or clear all	ON/AC	#5
Opening bracket (	(	#18
OR (bitwise)	OR	#6 → #22
P → R — polar to rectangular	P▶R	#1 → #39
Percent %	%	#6 → #15
Permutations nPr	nPr	#6 → #27
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	yx	#15
r — correlation coefficient	r	#1 → #35
R → P — rectangular to polar	R▶P	#6 → #39
RCL — recall a memory / variable	RCL	#26
Reciprocal x^{-1} (1/x)	1/x	#12
Round to the displayed precision	RND	#1 → #15
SCI display mode	SCI	#1 → #18
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	2nd	#6
Sine (uses the current angle unit)	SIN	#7
Square root √	√x	#14
Square x^2	x^2	#13
STAT1 — one-variable statistics	STAT1	#1 → #11
STAT2 — two-variable statistics	STAT2	#1 → #12

STO — store the value into a memory / variable (press the target next)	STO	#31
Subtraction −	−	#30
Sx — sample standard deviation	Sx	#6 → #28
Tangent (current angle unit)	TAN	#9
Unit conversion: multiply by 0.2641720524	▶gal	#1 → #33
Unit conversion: multiply by 0.3937007874	▶in	#1 → #32
Unit conversion: multiply by 0.45359237	▶kg	#6 → #34
Unit conversion: multiply by 2.2046226218	▶lb	#1 → #34
Unit conversion: multiply by 2.54	▶cm	#6 → #32
Unit conversion: multiply by 3.785411784	▶L	#6 → #33
$\bar{x}$ — mean	$\bar{x}$	#6 → #16
x-th root of y	x√y	#6 → #4
x≷y — exchange X and Y	x⇄y	#36
XNOR (bitwise)	XNOR	#6 → #24
XOR (bitwise)	XOR	#6 → #23
$\bar{y}$ — mean of y	$\bar{y}$	#1 → #16
π (3.14159265…) as an exact constant	π	#1 → #9
π (3.14159265…) as an exact constant	π	#16
σ — population standard deviation	σx	#1 → #28
Σ+ — enter a statistics data point	Σ+	#11

Σ− — remove a statistics data point	**Σ−**	**#6 → #11**
Σx — sum of x	**Σx**	**#6 → #29**
Σx^2 — sum of squares	**Σx^2**	**#1 → #29**

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

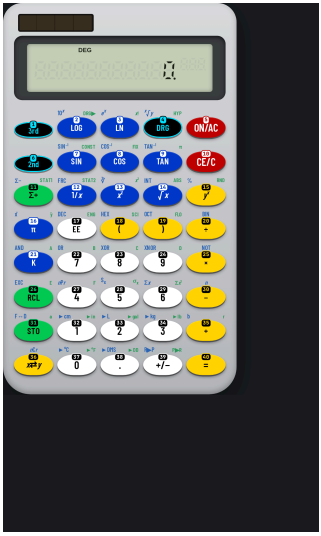
One-line algebraic (AOS) model with 2nd and 3rd shifts. STAT1/STAT2 (3rd Σ+/1/x) start 1- and 2-variable statistics; Σ+ enters data (x⇄y separates x and y). DEC/HEX/OCT/BIN base modes with AND/OR/XOR/XNOR/NOT (2nd), hex digits A–F on 3rd. Unit conversions ►cm/►in etc. on 1/2/3/0. Several unreadable labels were completed with this family's standard functions.

Planned (not yet working)

- Two-line display preview
- Fraction simplify step
- Complex mode
- Regression ŷ / x̂ prediction
- Physical constants list
- Answer history export
- Speech read-out of results
- Colour-blind key theme
- Exam lock
- Custom shortcut keys

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Appendix E — Quick-reference card



#1 3rd	#2 LOG ⇧ 10x	#3 LN ⇧ ex
#4 DRG ⇧ x√y	#5 ON/AC	#6 2nd
#7 SIN ⇧ SIN⁻¹	#8 COS ⇧ COS⁻¹	#9 TAN ⇧ TAN⁻¹
#10 CE/C	#11 Σ+ ⇧ Σ−	#12 1/x ⇧ FRC
#13 x² ⇧ ∛	#14 √x ⇧ INT	#15 yx ⇧ %
#16 π ⇧ x̄	#17 EE ⇧ DEC	#18 (⇧ HEX
#19) ⇧ OCT	#20 ÷ ⇧ BIN	#21 K ⇧ AND
#22 7 ⇧ OR	#23 8 ⇧ XOR	#24 9 ⇧ XNOR
#25 × ⇧ NOT	#26 RCL ⇧ EXC	#27 4 ⇧ nPr
#28 5 ⇧ Sx	#29 6 ⇧ Σx	#30 − ⇧ n
#31 STO ⇧ F↔D	#32 1 ⇧ ▶cm	#33 2 ⇧ ▶L
#34 3 ⇧ ▶kg	#35 + ⇧ b	#36 x⇄y ⇧ nCr
#37 0 ⇧ ▶℃	#38 . ⇧ ▶DMS	#39 +/− ⇧ R▶P
#40 =		

U-45 CALCULATORS · Calculator #12 · cut out and keep